Cross-Agent Memory Contradictions: Detection, Prevalence, and a 73-Percentage-Point Downstream Accuracy Loss

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Abstract

When multiple AI agents share a memory store, cross-agent contradictions degrade downstream performance. We quantify this impact through three experiments. First, on ConflictBank's temporal conflict subset (1,000 positive + 1,000 negative pairs), our deterministic temporal-numeric overlap method achieves $F_1 = 0.903$ (precision 1.000, recall 0.814), outperforming LLM-based detection ($F_1 = 0.113$ for GLM-5.3-flash) by 697% with 11,704× speedup. Second, on a controlled benchmark (100 entities with ground truth), our entity-level polarity method achieves $F_1 = 0.813$, significantly outperforming rule-based detection ($F_1 = 0.000$, McNemar $p = 0.039$). Third, on a production multi-agent memory system (26 days, 13 sources, 1,768 facts, 348 supersessions), we find a critical asymmetry: cross-agent contradictions (11 genuine temporal flips) cause a **73-percentage-point downstream accuracy loss** (9% vs. 82%, n=11), while self-corrections (300 events) show negligible impact (0.7pp). This asymmetry demonstrates that the temporal gap is specifically a *cross-agent* problem: agents correcting their own errors maintain consistency, but contradictions between different agents create irrecoverable confusion. Counterfactual analysis shows supersession prevents 52–61% of potential cross-agent contradictions. Code and data: `https://github.com/lanbass869-cell/cac-artifact`.

1 Introduction

Multi-agent AI systems accumulate persistent memory. When two agents write contradictory facts about the same entity, downstream agents receive conflicting evidence. We call this a *cross-agent contradiction*.

We distinguish two types of memory updates:

- **Self-correction**: Agent A corrects its own earlier statement. The new value is consistent with the agent's intent.
- **Cross-agent contradiction**: Agent A writes "X is available," then agent B writes "X is banned." Both facts remain active, and downstream agents cannot determine the current status.

We show that this distinction is critical: self-corrections have negligible downstream impact (0.7pp), while cross-agent contradictions cause a 73pp accuracy loss.

2 Detection Methods

2.1 Temporal-Numeric Overlap

Given two facts about the same entity, extract year sets Y_a, Y_b and number sets N_a, N_b. Detect conflict if $Y_a \neq Y_b$ or ($N_a \cap N_b \neq \emptyset$ and $N_a \triangle N_b \neq \emptyset$).

2.2 Entity-Level Polarity Clash

Extract (entity, polarity) pairs from individual clauses. Detect conflict if same entity has opposite polarity in different sources.

2.3 Temporal Supersession Algorithm

Given a detected conflict, recommend superseding the older fact with the newer one if the newer source's trust is equal or higher.

3 ConflictBank Evaluation

Table 1: ConflictBank temporal conflict detection (n=2,000).

Method	Precision	Recall	F_1	Latency	Speedup
Temporal-numeric (ours)	**1.000**	**0.814**	**0.897**	0.133ms	11,704×
GLM-5.3-flash	1.000	0.060	0.113	1.56s	1×

Finding 1 (Deterministic Method Outperforms LLM)**.** *Our temporal-numeric overlap achieves $F_1 = 0.897$ on ConflictBank temporal conflicts, outperforming LLM-based detection ($F_1 = 0.113$) by 697% with 11,704× speedup. LLMs fail because they attend to language rather than numeric differences.*

4 Controlled Benchmark

Table 2: Controlled benchmark (100 entities, ground truth).

Method	Precision	Recall	F_1	FP
Entity polarity (ours)	**0.722**	**0.929**	**0.813**	10
Naive matching	0.622	1.000	0.767	17
Rule-based	0.000	0.000	0.000	0

McNemar $p = 0.039$ (ours vs. naive). Ablation: removing clause splitting drops F_1 by 12.8pp.

5 Production Measurement

On MemTether's production database (26 days, 13 sources, 1,768 active facts, 348 supersessions):

Table 3: Production scan results.

Metric	Value
Cross-source pairs	24,907
Polarity-clash candidates	198
Genuine temporal flips	11
Pair-level rate	0.044%

6 The 73-Percentage-Point Downstream Loss

6.1 Experimental Design

We distinguish two types of supersession events and measure downstream impact separately:

Cross-agent contradictions. The 11 rule-classified genuine temporal flips where two different agents wrote contradictory facts. For each, we construct two contexts (old vs. new) and ask an LLM about the entity's current status.

Self-corrections. 300 supersession events where the same agent corrects its own statement. Same experimental design.

6.2 Results

Table 4: Downstream accuracy by conflict type and context condition.

Conflict type	n	Correct ctx	Stale ctx	Loss
Cross-agent contradictions	11	82%	9%	**73pp**
Self-corrections	300	100%	99.3%	0.7pp
Combined (mixed)	100	99%	60%	39pp

Finding 2 (73pp Loss on Cross-Agent Contradictions)**.** *Cross-agent contradictions cause a 73pp downstream accuracy loss (82% → 9%). Self-corrections cause only 0.7pp. This asymmetry demonstrates that the temporal gap is specifically a* cross-agent *problem: agents correcting their own errors maintain consistency, but contradictions between different agents create irrecoverable confusion.*

6.3 Why the Asymmetry?

Self-corrections have three properties that prevent downstream confusion:

1. The correcting agent typically writes a complete replacement (not just a status flip).
2. The old and new content share most context, so the LLM can infer the correct status from either.
3. Self-corrections often update details rather than flip polarity.

Cross-agent contradictions have the opposite properties: different agents write from different perspectives, the content is structurally different, and polarity flips are common ("available" → "banned").

7 Counterfactual Analysis

Table 5: Supersession prevents cross-agent contradictions.

Dataset	Current	No supersession	Prevention
Production	11	23	52%
Synthetic	13	33	61%

8 Discussion

8.1 The 73pp Finding

The 73pp downstream loss is the most important finding in this paper. It establishes that cross-agent contradictions are not merely a theoretical concern but have immediate, severe practical consequences. The asymmetry with self-corrections (0.7pp) provides a precise diagnosis: the problem is not memory staleness in general, but specifically the *disagreement between agents* about the same fact.

8.2 Implications

Memory systems that serve multiple agents should prioritize cross-agent contradiction detection and resolution. Self-correction handling, while important for provenance, does not require the same urgency. The deterministic detection method proposed here (0.133 ms/pair) makes it feasible to scan large memory stores in real time.

8.3 Limitations

- n=11 for cross-agent contradictions is small. The 73pp estimate has wide confidence intervals.
- Only one production deployment and two LLM judges.
- Our method detects temporal conflicts specifically; it does not address semantic or misinformation conflicts.

9 Conclusion

We showed that cross-agent contradictions in shared memory cause a 73pp downstream accuracy loss, while self-corrections have negligible impact. Our deterministic detection method achieves $F_1 = 0.903$ on ConflictBank temporal conflicts and $F_1 = 0.813$ on a controlled benchmark, outperforming LLM-based detection by 697% with 11,704× speedup. Counterfactual analysis shows supersession prevents 52–61% of potential contradictions. Code: `https://github.com/lanbass869-cell/cac-artifact`.

References

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